

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2049

North Blacksburg 138 kV
June 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2049. The Interconnection Customer proposes a 165.3 MW solar photovoltaic facility interconnecting at the North Blacksburg 138 kV substation in the Kentucky area. The request was received on April 01, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 665 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,089,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Sierra Pine – Arpgen 138kV
Contingency: Pine Bluff Watson Chapel – Carraizo 1201 345kV
Loading: 143.0% (Rating: 291 MVA)
DFAX: 100.0%
Pre-project loading: 86.2%
2. Facility: Arpgen – N Blacksburg 138kV
Contingency: Carraizo 1201 – Pine Bluff Watson Chapel 345kV
Loading: 140.6% (Rating: 306 MVA)
DFAX: 100.0%
Pre-project loading: 86.6%
3. Facility: Sierra Pine – Schertz 138kV
Contingency: Sierra Pine Substation – Carntown 138kV line
Loading: 135.7% (Rating: 241 MVA)
DFAX: 74.6%
Pre-project loading: 84.5%
4. Facility: N Selma – W Olean 138kV line
Contingency: Arverne – Deschutes 138kV
Loading: 132.7% (Rating: 193 MVA)
DFAX: 46.8%
Pre-project loading: 92.6%
5. Facility: Sierra Pine – Carntown 138kV Line
Contingency: Schertz – Sierra Pine 138kV L
Loading: 132.7% (Rating: 227 MVA)
DFAX: 64.3%
Pre-project loading: 85.9%

6. Facility: Carntown Substation – Cmf Vaaville 138kV
Contingency: Schertz – Sierra Pine 138kV L
Loading: 131.0% (Rating: 226 MVA)
DFAX: 64.3%
Pre-project loading: 84.0%
7. Facility: Cmf Vacaville – North Selma 138kV
Contingency: Sierra Pine – Schertz 138kV
Loading: 129.6% (Rating: 219 MVA)
DFAX: 64.3%
Pre-project loading: 81.2%
8. Facility: Schertz – Met Ctr 138kV
Contingency: Sierra Pine – Carntown 138kV line
Loading: 117.5% (Rating: 234 MVA)
DFAX: 38.9%
Pre-project loading: 90.0%
9. Facility: Arverne Substation – Tyranaena 138kV
Contingency: Beaver DamEast – Keifer-Jackson 138kV Line
Loading: 116.8% (Rating: 201 MVA)
DFAX: 15.7%
Pre-project loading: 103.9%
10. Facility: Dynergy Arlington Vly Energy Facility – Met Center 138kV
Contingency: Sierra Pine Substation – Carntown 138kV line
Loading: 115.7% (Rating: 229 MVA)
DFAX: 38.9%
Pre-project loading: 87.6%
11. Facility: N Selma – Dynergy Arlington Vly Energy Facility 138kV
Contingency: Sierra Pine – Carntown 138kV
Loading: 113.6% (Rating: 226 MVA)
DFAX: 38.9%
Pre-project loading: 85.1%
12. Facility: Lg&E; Canal – Tyranaena Substation 138kV
Contingency: Keifer-Jackson – Beaver Dam East 138kV
Loading: 112.8% (Rating: 219 MVA)
DFAX: 8.9%
Pre-project loading: 106.1%
13. Facility: Eldrado 345 kV/138 kV Transformer
Contingency: Carraizo 1201 – Pine Bluff Watson Chapel 345kV
Loading: 107.1% (Rating: 544 MVA)
DFAX: 21.1%
Pre-project loading: 100.7%
14. Facility: Deschutes – Arverne 138kV
Contingency: W Olean – N Selma 138kV Line
Loading: 106.6% (Rating: 198 MVA)
DFAX: 31.8%
Pre-project loading: 79.9%
15. Facility: Cairo – Becker 345kV
Contingency: Dillard-Orange – Beck Rd Substation 345kV Line
Loading: 103.6% (Rating: 1,379 MVA)
DFAX: 6.1%
Pre-project loading: 102.8%
16. Facility: Hopson – L&E; Canal 100kV Line 1
Contingency: Lg&E; Canal – Hopson Substation 100kV

Loading: 102.0% (Rating: 203 MVA)

DFAX: 6.2%

Pre-project loading: 96.9%

17. Facility: Hopson – Lg&E; Canal Substation 100kV Ckt 2

Contingency: Lg&E; Canal – Hopson 100kV Cir. 1

Loading: 102.0% (Rating: 203 MVA)

DFAX: 6.2%

Pre-project loading: 96.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sierra Pine – Arpgen 138kV	Rebuild line with higher-capacity conductor	\$16,555,000	50% (\$8,312,000)
North Blacksburg – Arpgen 138kV	Rebuild line with higher-capacity conductor	\$25,113,000	50% (\$12,610,000)
Schertz – Sierra Pine 138kV Line	Rebuild line with higher-capacity conductor	\$18,023,000	48% (\$8,749,000)
N Selma – W Olean 138kV line	Rebuild line with higher-capacity conductor	\$5,225,000	28% (\$1,451,000)
Sierra Pine – Carntown 138kV Line	Rebuild line with higher-capacity conductor	\$24,800,000	48% (\$12,010,000)
Cmf Vacaville – Carntown 138kV	Rebuild line with higher-capacity conductor	\$6,980,000	48% (\$3,380,000)
Cmf Vacaville – North Selma 138kV	Rebuild line with higher-capacity conductor	\$3,058,000	48% (\$1,481,000)
Met Ctr – Schertz 138kV	Rebuild line with higher-capacity conductor	\$2,772,000	28% (\$788,000)
Arverne Substation – Tyranena 138kV	Rebuild line with higher-capacity conductor	\$2,332,000	13% (\$294,000)
Dynergy Arlington Vly Energy Facility – Met Center 138kV	Rebuild line with higher-capacity conductor	\$10,536,000	28% (\$2,994,000)
N Selma – Dynergy Arlington Vly Energy Facility 138kV	Rebuild line with higher-capacity conductor	\$4,750,000	28% (\$1,350,000)
Lg&E; Canal – Tyranena 138kV L	Rebuild line with higher-capacity conductor	\$2,180,000	28% (\$613,000)
Eldorado 345-138 kV Auto	Replace transformer with higher-rated unit	\$15,025,000	6% (\$931,000)
Arverne – Deschutes 138kV	Reconductor line	\$2,128,000	44% (\$924,000)
Cairo – Becker 345kV	Reconductor line	\$10,715,000	1% (\$128,000)
Hopson – L&E; Canal 100kV Line 1	Reconductor line	\$106,000	29% (\$31,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hopson – Lg&E; Canal 100kV line	Reconductor line	\$147,000	29% (\$43,000)
	Total allocated to project		\$56,089,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.